

HAVANA CASABLANCA, Cuba

WMO: 783250

Lat: 23.144N

Lon: 82.342W

Elev: 164

StdP: 14.61

Time Zone: -5.00 (CUB)

Period: 90-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	60.7	63.0	45.8	45.7	65.1	49.3	52.4	66.4	24.9	73.9	23.0	72.8	7.3	30	0.332

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.6	90.0	77.5	88.9	77.5	88.0	77.6	80.1	86.5	79.5	85.9	79.0	85.5	10.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.2	147.6	83.9	77.6	144.4	83.4	77.1	142.2	83.0	43.7	86.8	43.0	86.2	42.6	85.9	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	20.4	18.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.7	72.8	73.2	73.6	77.0	79.2	81.3	82.0	82.5	81.6	79.7	76.1	73.2
	DBStd	4.63	3.61	3.89	4.00	2.90	2.07	1.87	1.44	1.55	1.75	2.40	2.97	3.67
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	7	2	2	1	0	0	0	0	0	0	0	0	2
	CDD50	10115	707	649	731	811	905	938	993	1009	949	921	783	719
	CDD65	4648	244	231	267	361	440	488	528	544	499	456	333	257
	CDH74	38167	925	1078	1398	2492	3618	4872	5687	6023	5070	3995	1964	1045
	CDH80	10264	96	168	240	520	835	1464	1920	2134	1566	924	293	104
(14) Wind	WSAvg	7.7	9.0	9.6	9.7	9.7	8.2	7.5	5.7	5.3	5.1	6.6	7.4	8.4
(15) Precipitation	PrecAvg	46.3	3.0	2.6	1.6	2.6	3.8	6.7	4.2	4.3	5.8	6.4	3.6	2.6
	PrecMax	74.2	12.2	8.9	9.3	8.3	13.0	27.3	11.1	10.4	16.6	20.6	16.8	8.9
	PrecMin	26.1	0.0	0.1	0.0	0.0	0.0	0.5	0.7	0.6	1.1	0.2	0.2	0.0
	PrecStd	11.3	2.7	1.9	1.6	2.2	3.1	5.0	2.7	2.2	3.3	4.1	3.0	2.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.0	87.7	88.4	89.9	89.5	91.5	90.3	91.8	91.9	89.9	87.9	85.6
		MCWB	72.6	72.2	73.1	74.9	77.6	78.8	77.6	78.1	78.0	77.4	76.4	73.7
	2%	DB	82.6	84.6	85.5	86.8	86.7	88.9	88.7	89.2	89.0	87.3	84.6	82.6
		MCWB	72.1	72.5	73.1	74.4	75.9	78.1	77.8	78.2	77.7	77.8	75.0	73.0
	5%	DB	80.4	81.6	82.5	84.5	85.4	87.3	87.7	88.4	87.7	85.6	82.9	80.4
		MCWB	72.1	72.0	72.4	73.6	75.4	78.0	78.0	78.2	77.7	77.4	74.8	72.5
	10%	DB	78.7	79.5	80.4	83.0	84.2	86.2	86.9	87.5	86.6	84.3	81.4	78.9
		MCWB	71.5	71.5	71.4	73.0	74.7	77.7	77.9	78.1	77.5	76.7	74.0	71.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.4	75.5	76.1	77.4	79.3	80.8	80.7	80.9	80.0	80.4	77.6	75.9
		MCDB	80.9	81.4	83.2	84.8	85.9	86.9	86.8	87.2	86.6	85.8	83.1	81.0
	2%	WB	74.3	74.2	74.9	76.1	78.1	79.8	79.7	80.0	79.2	79.1	76.9	74.7
		MCDB	79.5	79.8	80.8	83.2	84.2	86.0	86.2	86.7	85.9	84.7	82.6	79.5
	5%	WB	73.4	73.4	74.1	75.1	77.0	79.1	79.0	79.3	78.7	78.3	76.1	73.8
		MCDB	78.3	78.9	80.0	81.8	83.0	85.2	85.7	85.9	85.5	83.7	81.2	78.8
	10%	WB	72.5	72.6	73.1	74.3	76.1	78.4	78.5	78.8	78.2	77.6	75.2	72.8
		MCDB	77.2	77.8	79.0	80.9	82.0	84.4	85.0	85.5	84.7	82.8	79.6	77.7
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	11.7	11.6	11.2	10.2	10.1	10.3	10.6	10.4	9.3	9.4	10.0
		MCDBR	13.9	15.2	14.9	13.7	11.7	11.6	11.1	11.6	12.0	11.2	12.1	12.9
	5% WB	MCWBR	5.6	6.1	6.1	5.0	5.1	4.9	4.8	4.5	4.4	4.3	4.6	5.6
		MCWBR	12.0	13.3	12.6	12.4	10.6	10.7	10.5	10.8	11.3	10.4	11.0	11.3
(40) Clear-Sky Solar Irradiance	taub		0.362	0.370	0.378	0.401	0.426	0.447	0.454	0.431	0.416	0.407	0.377	0.372
		taud	2.528	2.511	2.472	2.410	2.380	2.377	2.347	2.434	2.467	2.471	2.505	2.521
	Ebn at Noon		287	291	293	286	276	268	266	274	278	276	279	279
		Edh at Noon	30	33	35	38	39	39	40	37	35	33	31	30
(44) All-Sky Solar Radiation	RadAvg		1237	1512	1842	2064	2036	1897	2051	1958	1724	1484	1283	1174
	RadStd		99	105	68	72	193	139	94	68	89	106	61	95

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	+1.60	N/A	N/A	N/A	N/A	N/A	-20	N/A	N/A

Nomenclature: See separate page